

THE ACTUATOR PETTING ZOO

Hands-on multimodal prototyping · Week 39 lab · 2 hours

WHY WE ARE HERE

You have had the actuator lecture. You know what an ERM is on a slide.

Today you put one against your own wrist, decide it feels cheap, and pick something else — before you commit a project to it.

Every year projects arrive at the mid-way pitch built entirely on the screen and the speaker.

Not because haptics was the wrong choice — because nobody knew a coin motor costs 12 kr and takes two wires.

HOW TODAY WORKS

Six benches. Already wired. Already running.

- Ten minutes each, on a hard timer.
- You change one parameter and answer one question.
- You wire nothing — that is what the project weeks are for.

Then you build one tacton and someone else tries to read it.

RUN SHEET

Times are offsets from the start. The rotation is the spine — keep it moving.

0:00	10 min	Framing, and a demo that fails
0:10	60 min	Bench rotation — 10 minutes each
1:10	10 min	Round-the-room: what surprised you
1:20	30 min	Build one tacton
1:50	10 min	Blind test and pack down

THE SIX BENCHES

01 — ERM COIN MOTOR

CUTANEOUS · VIBRATION

The eccentric rotating mass from Lecture 6. The same part that is in your phone and every game controller.

Turn the pot and notice what you cannot do: frequency and amplitude are mechanically welded together. "Gentle but fast" is not available to you.

part 10mm 3V coin ERM
drive 2N2222 + 1N4148
pin D9 (PWM)
draw ~75 mA
spin-up 20–40 ms
cost ~12 kr

CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?

02 — LRA + PIEZO DISC

CUTANEOUS · VIBRATION

Two ways out of the ERM's compromise.

The LRA hits resonance in ~5 ms and stops just as fast — which is why phone keyboards use it.

The piezo is the crystal from the lecture: near-instant, almost no power, but shallow. A tick, not a thump.

parts LRA 10mm + piezo
driver DRV2605L (I2C)
pin A4 SDA / A5 SCL
effects 123 waveforms
cost ~90 kr

CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why not the others?

03 — SOLENOID TAP

CUTANEOUS · IMPACT

A push-pull solenoid firing a single 15 ms pulse against a fingertip.

This is the bench everyone remembers. A discrete knock carries urgency no amount of buzzing does — it reads as somebody tapping you.

Indexical, not symbolic.

part	5V push-pull
drive	MOSFET + flyback
pin	D6
peak	~1.1 A
duty	<= 25%, gets hot
cost	~45 kr

CARD QUESTION

How many taps before it goes from alert to nagging?

04 — CAPACITIVE TOUCH

KINESTHETIC · TOUCH-TO-ACTUATE

Touch a surface with a bare finger and the servos react.

There is no button and no sensor you can point at. The input IS a wire taped to a plate, read on an analogue pin.

This is the SOS 2025 rig running its real sketch.

```
sense    ADCTouch on A0
baseline  25-sample roll
trigger   1.01x average
debounce  100 ms
servos    D4 + D13
cost      ~0 kr – a wire
```

CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?

BENCH 04 — THE WHOLE TRICK

Capacitive sensing costs a wire. What it does not cost you is thinking.

reading > baselineAverage() * 1.01

The threshold is RELATIVE, never absolute.

- Readings drift with humidity, mains hum, how you are sitting. A fixed threshold works in the morning and fails after lunch.

The baseline only learns while untouched.

- Otherwise a long press slowly teaches it that a finger is normal, and the touch silently stops registering.

05 — PELTIER WARM / COOL

CUTANEOUS · THERMAL

A thermoelectric tile: heats one side, cools the other, reverses when you flip the current.

Sit with it. It takes several seconds before you are sure which way it went.

The lecture's warning holds up in person — alternating hot and cold just reads as lukewarm.

part TEC1-12706
drive L298N H-bridge
pin D3 PWM / D4-5
draw 2–4 A, bench PSU
onset 3–8 s
cap 45 C in software

CARD QUESTION

Time yourself. How long until you would bet money on warmer vs cooler?

06 — THE SAME SIGNAL, TWICE

INTRAMODAL · AUDIO + HAPTIC

One transducer on a plate, playing 40 Hz you feel and 400 Hz you hear
— from the same driver.

Toggle between them alone and together.

Together is not louder. It is more CERTAIN. That is redundancy from
Lecture 7, and the cheapest reliability win your project can get.

part	bone-conduction
amp	PAM8403 class-D
pin	D11 (tone)
felt	~40 Hz
heard	~400 Hz
cost	~110 kr

CARD QUESTION

With ear defenders on, does the tacton still work?

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch — a wire on an analogue pin. See bench 04.
- Heart rate (PPG) — steady at rest, garbage the moment the hand moves.
- GSR — arousal, but it drifts. Relative change only, never absolute.
- Flex sensor — the cheap route to a data glove.
- Pressure (FSR) — how hard, not just whether. Calibrate per pad.
- Your own phone — accelerometer, gyro, mic, camera, vibration. No soldering at all.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Capacitive sensing + servos + actuators — SOS 2025, Arduino Uno

- A moving screen that breathes when idle and retreats when touched. 922 lines, plus a browser control panel over serial.
- Take away: easing and idle motion are what separate "a servo moved" from "it reacted to me". Both are a dozen lines.

Instrumented sock — bachelor project, ESP32

- Six force sensors under a foot, batched to a server over Wi-Fi, each with its own calibration constants.
- Take away: calibrate per sensor, and batch your writes.

BUILD ONE TACTON

THE BRIEF

30 minutes. Deliberately not long enough to be precious about it.

Pick one actuator from the rotation.

Encode three messages: arrived / something wrong / finished.

Vary RHYTHM and ROUGHNESS only — the Lecture 7 tacton parameters. No swapping actuators between messages.

Hand it to another group with the display turned away. They name all three.

Write down which two got confused, and what would have separated them.

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If your design only works when someone is watching a screen, you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop